MIL-P-80085B
16 April 1985
SUPERSEDING
MIL-P-80085A
27 May 1970

MILITARY SPECIFICATION

POSITIONERS, WELDING, HEADSTOCK, TAILSTOCK, AND COMBINATION TYPES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers power driven headstock type positioners, non-powered idler tailstock type positioners, combination headstock-tailstock positioners and combination tailstock-tailstock positioners with the function of rotating workpieces while welding operations are performed.

1.2 Classification. This specification covers welding positioners of the following types and sizes. The type and size to be furnished shall be as specified (see 6.2.1).

Type I - Headstock type positioner.
Type II - Tailstock type positioner.
Type III - Headstock-tailstock type positioner.
Type IV - Tailstock-tailstock type positioner.

Size 1 - 2,500 lb. (1134 kg.) capacity.
Size 2 - 3,000 lb. (1361 kg.) capacity.
Size 3 - 4,000 lb. (1814 kg.) capacity.
Size 4 - 5,000 lb. (2268 kg.) capacity.
Size 5 - 7,500 lb. (3402 kg.) capacity.
Size 6 - 10,000 lb. (4536 kg.) capacity.
Size 7 - 12,500 lb. (5670 kg.) capacity.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvment Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3436

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Size 8 - 15,000 lb. (6804 kg.) capacity.
Size 9 - 20,000 lb. (9072 kg.) capacity.
Size 10 - 24,000 lb. (10887 kg.) capacity.
Size 11 - 30,000 lb. (13608 kg.) capacity.
Size 12 - 37,000 lb. (16783 kg.) capacity.
Size 13 - 50,000 lb. (22680 kg.) capacity.
Size 14 - 60,000 lb. (27216 kg.) capacity.
Size 15 - 90,000 lb. (40824 kg.) capacity.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units For General Use by the Federal Government.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issue of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NEMA MG 1 - Motors and Generators.
ANSI/NEMA ICS 1 - Industrial Control and Systems.
ANSI/NFPA 79 - Metalworking Machine Tools and Plastics Machinery, Electrical Standards for.
ANSI B5.1 - T-Slots, Their Bolts, Nuts and Tongues.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First Article. When specified (see 6.2.1), a sample positioner shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Design. The positioners shall be new and one of the manufacturer's current models capable of supporting, rotating, and positioning workpieces for effective welding operations. The positioners shall include all components, parts, and features necessary to meet the performance requirements specified herein. The positioners shall be designed with a base and frame, horizontal shaft or rotating spindle, and a vertical face table mounted on the spindle. Additional features which are not specified herein that are a part of the manufacturer's current design shall be included in the positioners furnished. The headstock type and the tailstock type of the same size shall be designed with similar or matching dimensions, except the headstock

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type shall be designed with electric motor driven table rotation. The positioner design shall be such as to provide convenient, efficient, and safe operation. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair. The positioners and component parts shall be designed to operate satisfactorily in ambient temperatures up to 120 degrees Fahrenheit. All units shall have a self-locking device to prevent inadvertent rotation of the table. Positioner parts and components subject to contamination from flux and foreign matter shall be enclosed or shielded to prevent contamination.

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the positioner. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as scales, depth stops, carriage stops, dial indicators, pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the positioners shall be graduated in the same system.

3.2.2 Reclaimed materials. The positioner may contain reclaimed materials provided such materials will not jeopardize the positioners intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the positioner.

3.2.3 Energy efficiency. The headstock and headstock-tailstock positioners and all applicable components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the Industry.

3.2.4 Controls. All operating controls shall be located convenient to the operator at his normal work station.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the positioner that present safety hazards. The safety devices shall not interfere with the operation of the positioner. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, mainte-

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nance, and repair of the parts. All positioner parts, components, mechanisms, and assemblies furnished on the positioner, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the positioner itself. Additional safety and health requirements shall be as specified (see 6.2.1).

3.2.6 Mercury restriction. The postioners shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the positioners.

3.2.8 Environmental protection. The positioners shall be so designed and constructed that, under the operating, service, transportation, and storage conditions described herein, the positioners shall not emit materials hazardous to the ecological system as prescribed by federal, state, or local statutes in effect at point of installation.

3.2.9 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.10 Interchangeability. All parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or positioner.

3.3 Construction. The positioners shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects such as scale and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the positioner. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

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3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the positioners shall be painted in accordance with the manufacturer's commercial practice, provided that all surfaces to be painted shall be cleaned of all foreign matter and at least one coat of primer and one coat of finish color is applied.

3.3.6 Threads. All threaded parts used on the positioners and related attachments and accessories shall conform to FED-STD-H28.

3.3.7 Dials. All dials that indicate positioner movement or rotational speed shall be such that graduations may be read from the normal operating position. Dials shall be permanently and legibly engraved or etched on a nonglare background.

3.3.8 Gears. All gears shall be constructed from a material suitable for the intended purpose. All gears shall be machined and heat treated by a process that will impart the hardness, surface finish, and toughness that will enable the gear train to transmit full rated torque of the drive motor without gear damage, failure, or premature deterioration and wear.

3.4 Components.

3.4.1 Type I positioners. The type I positioner shall consist of at least a frame and base assembly, a horizontal rotating spindle supporting a faceplate or a work table with a vertical face, and an electric motor drive and controls for table rotation.

3.4.2 Type II positioners. The type II positioner shall be similar to the type I positioner except the unit shall have no electric motor drive or controls.

3.4.3 Type III positioners. The type III positioner shall consist of one headstock unit and a matching tailstock unit, with provisions for in-line mounting and operation similar to a lathe. The units shall be mounted on a common surface or rails.

3.4.4 Type IV positioners. The type IV positioner shall consist of two matched tailstock units with provisions for in-line operation similar to a lathe. The units shall be mounted on a common surface or rails.

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3.4.5 Frame and base. The frame and base assembly shall be a steel weldment construction, ribbed to maintain original alignment and to minimize deflection. The base shall be provided for floor or platform mounting. When optional sub-base frames are required, they shall be as specified (see 6.2.1).

3.4.6 Table. The table shall be of sufficient strength to maintain alignment for performance of circular welding at capacity load conditions. Unless otherwise specified, the table work surface shall be flat to within 3/32 of an inch. When required, the table face shall be machined as specified (see 6.2.1). Unless otherwise specified (see 6.2.1), the table shall be square, and shall be not less than the dimensions specified in table I. Table center height shall be not less than the height shown in table I.

3.4.6.1 Table T-slots. Unless otherwise specified (see 6.2.1), each table shall be furnished with four equally spaced radial T-slots. The T-slots shall conform to ANSI B5.1 and shall be the self-cleaning type. The T-slot length shall be not less than 80 percent of the diagonal dimension of the table. Four matching T-bolts and nuts shall be furnished for each table slot.

3.4.6.2 Table spindle. Unless otherwise specified (see 6.2.1), a hollow spindle shall be furnished on all positioners. Spindle bearings shall be as required for continuous performance and shall be protected from welding current.

3.4.6.3 Table rotation drive. The headstock and headstock-tailstock type positioners shall be equipped with electric motor driven table rotation on the headstock only. The table rotation shall be accomplished by an electric motor with spindle rotational torque not less than that specified in table I. Unless otherwise specified (see 6.2.1), table speed shall be constant and shall be not less than that specified in table I. Deviation of constant table speed from control set point shall be not greater than ± 5 percent. The table shall also have a variable speed range conforming to table I, with deviations of no more than 25 percent. When specified (see 6.2.1), a high speed table rotation drive shall be furnished with speeds as specified. Adjustment shall be provided to eliminate all backlash in the table due to wear.

3.4.7 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to ANSI/NFPA 79. Each headstock unit shall draw all of its electrical power from a single 230/460 volt, 3 phase, 60 Hz circuit, and shall have a fused safety disconnect switch or circuit breaker. Each headstock unit shall be initially wired for operation on 230 volts. An identified terminal for grounding the positioner when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in ANSI/NFPA 79 for the disconnect fuse rating.

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TABLE I. Sizes and capacities.
Types I, II, III, and IV positioners.

Capacities	Positioner size														
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Unit load capacity	2,500	3,000	4,000	5,000	7,500	10,000	12,500	15,000	20,000	30,000	30,000	37,000	50,000	60,000	90,000
Table size (inches per side of sq. table)	30	30	30	40	45	48	48	54	54	54	54	54	54	54	54
Table center to floor (inches)	27	27	27	36	36	36	36	50	50	50	50	50	50	50	50
Table thickness (inches)	1-3/4	1-3/4	1-3/4	2	2	2	2	2	2	2-1/2	2-3/4	3	3	3	3
Type I and type III positioners only.															
Constant speed (RPM)	1.00	1.00	1.00	1.00	1.00	1.00	.50	.50	.50	.40	.40	.40	.40	.50	.50
Variable speed range (RPM)															
Minimum	.065	.065	.065	.070	.080	.065	.065	.070	.070	.070	.070	.040	.040	.050	.050
Maximum	.50	.50	.50	.50	.50	.50	.50	.50	.40	.30	.30	.30	.30	.30	.050
Spindle rotational torque (inch-lbs)	15,000	36,000	36,000	36,000	36,000	120,000	150,000	240,000	240,000	288,000	360,000	444,000	600,000	800,000	900,000
Type III and type IV positioners only.															
Load capacity between units (lbs)	5,000	6,000	8,000	10,000	15,000	20,000	25,000	30,000	40,000	48,000	60,000	74,000	100,000	120,000	180,000

All values shall be not less than as shown except that where a range is shown, the values shall be not more than the minimum and not less than the maximum figures shown.

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3.4.7.1 Magnetic reversing starter and safety switch. Unless otherwise specified (see 6.2.1), the headstock unit shall be furnished with a safety disconnect switch and magnetic reversing starter, both with overload protection. Electronic type speed drive controls shall be mounted in an enclosure conforming to ANSI/NEMA ICS 1, style 1.

3.4.7.2 Electric motors. Electric motors shall be rated for continuous duty, and shall be equipped with ball or roller bearings. Bearings shall be of the sealed and permanently lubricated type. Motor characteristics shall conform to ANSI/NEMA MG 1. Motor horsepower shall be as required to meet the performance requirements of table I. Motors shall be dual voltage type. Unless otherwise specified (see 6.2.1), motors shall be totally enclosed.

3.4.7.3 Headstock remote controls. Unless otherwise specified (see 6.2.1), all headstock units shall be equipped with a remote pendant type hand control having a flexible cable length of not less than fifteen feet. The pendant hand control shall have at least the following controls:

a. Power "on" pilot light

b. On-off switch or button

c. Forward or reverse selector

d. High speed control (when furnished)

3.4.7.4 Control circuit voltage. Auxiliary control circuits shall be isolated from the input electric power supply by a low-voltage transformer having a secondary voltage no greater than 115 volts.

3.4.7.5 Weld current grounding device. A weld current grounding device shall be furnished with each positioner. The capacity of the grounding device shall be not less than 2,000 amperes direct current (DC) with suitable frame-to-ground cable attachment. The attachment device for securing the welding ground cable to the positioner shall be permanently mounted in a position which will prevent interference during full operational ranges of the positioner. The type connection shall assure low resistance electrical conductivity.

3.5 Sizes and capacities. Unless otherwise specified (see 6.2.1), the sizes and capacities shall be not less than those specified in table I. Unit load capacity for type I and II units is the weight of a workpiece that the unit shall support with the center of gravity of the workpiece spaced 6 inches out from the table face. Load capacity for type III and type IV units is the weight of a workpiece that can be supported with the weight equally distributed between units.

3.6 Performance. The positioners shall be capable of rotational positioning of workpieces with performance capacities not less than those specified in table I. The positioners shall be capable of continuous operation and shall withstand the rough usage encountered by this type of equipment.

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3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.7 Standard equipment. Unless otherwise specified, one set of hand tools, normally furnished by the manufacturer shall be furnished.

3.8 Optional equipment. When required, the following optional equipment shall be furnished with each positioner as specified (see 6.2.1).

a. Round Table
b. Magnetic Clamps
c. Pneumatic Clamps
d. Tachometers
e. Foot Operated Controls
f. Interconnecting Rails

3.9 Repair parts. Such repair parts as are specified (see 6.2.1), shall be furnished.

3.10 Marking on plates and charts. All words on plates and charts shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

3.10.1 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each positioner. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.10.2 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each positioner. The nameplate shall contain the information listed below. If the positioner is a special model, the model designation shall include the model of the basic standard positioner and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

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Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract. All technical data shall be printed in the English language.

3.12 Workmanship. Workmanship of the positioner and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests in 4.6. Failure of the item to pass the examination and all tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.5, the test in 4.6.1, and the inspection in 4.7. Failure of the item to pass the examination, test and inspection shall be cause for rejection.

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4.5 Examination. The positioner and equipment shall be visually examined to determine compliance with the requirements of this specification.

4.6 Tests.

4.6.1 Performance test. The positioner shall be placed on a firm, level surface. A capacity load specified from table I shall be mounted on the table with the center of gravity located at the distance from the table specified in table I. The headstock units shall be connected to the specified source of electricity (see 3.4.7). All units shall be operated to determine compliance with 3.6. Positioner tables shall rotate freely with no evidence of galling or binding.

4.6.2 Weld current grounding test. The positioner shall have a full amperage welding load (see 3.4.7.5) passed through the ground circuit for 30 seconds with no damage or excessive heating of the positioner.

4.7 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Packaging, packing, and marking. Unless otherwise specified, packaging, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A or level B preservation, level A or level B packing and special marking shall be accomplished in accordance with MIL-W-45562.

6. NOTES

6.1 Intended use. Headstock and tailstock units and sets are used to rotate weldments for downhand welding.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and size required (see 1.2).

c. When first article is required for inspection and approval (see 3.1).

d. If machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

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e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Additional safety and health requirements, if required (see 3.2.5).

g. Painting, if different (see 3.3.5).

h. Sub-base frame, if required (see 3.4.5).

i. Machined table face, if different (see 3.4.6).

j. Table dimensions, if different (see 3.4.6).

k. T-slots, if different (see 3.4.6.1).

l. Table spindle, if different (see 3.4.6.2).

m. Table speed, if different (see 3.4.6.3).

n. Table high speed, if required (see 3.4.6.3).

o. Electrical system and input power supply, if different (see 3.4.7).

p. Magnetic reversing starter, and safety disconnect switch, if different (3.4.7.1).

q. Electric motor enclosure, if different (3.4.7.2).

r. Pendant control, if different (see 3.4.7.3).

s. Sizes and capacities, if different (see 3.5).

t. Additional performance features, if required (3.6.1).

u. Specify and fully describe optional equipment, if required (see 3.8).

v. Specify applicable repair parts, if required (see 3.9).

w. Lubrication chart or plate, if different (see 3.10.1).

x. Nameplate, if different (see 3.10.2).

y. First article inspection, if different (see 4.3).

z. Quality conformance inspection, if different (see 4.4).

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aa. If military packaging, packing, and marking in accordance with MIL-W-45562 is required, specify level required and specification number (see 5.1).

6.2.2 Contract data requirements. Required technical data such as operators manuals, parts lists, wiring diagrams, and examination, original alignment, and tolerance test reports, and other instructions for operation and maintenance (as identified on a DD Form 1664, Data Item Description) should be specified on a DD Form 1423, Contract Data Requirements List incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the positioner itself. It does not require compliance with those OSHA requirements that concern "the positioner in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat. Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual positioners in the environment) the procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional positioner requirements that will integrate the new positioner into its future operating environment. The positioner shall be equipped with all point-of-operation guarding normally furnished as standard on the manufacturer's commercial positioners supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required, as in most cases, the guard configuration is dependent on the size and configuration of the workpieces. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1 (f).

6.5 Cross reference of classification changes. Classification changes from previous issues of this specification are as follows:

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Size 1	Size 1 (no change)	Size 1 (no change)
	Size 2 (new size)	Size 2 (no change)
	Size 3 (new size)	Size 3 (no change)
Size 2	Size 4	Size 4 (no change)
	Size 5 (new size)	Size 5 (no change)
Size 3	Size 6	Size 6 (no change)
	Size 7 (new size)	Size 7 (no change)
Size 4	Size 8	Size 8 (no change)
Size 5	Size 9	Size 9 (no change)

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Size 6	deleted	
	Size 10 (new size)	Size 10 (no change)
Size 7	Size 11	Size 11 (no change)
	Size 12 (new size)	Size 12 (no change)
Size 8	deleted	
Size 9	Size 13	Size 13 (no change)
Size 10	deleted	
		Size 14 (new size)
		Size 15 (new size)
Size 11	deleted	
Size 12	deleted	
Size 13	deleted	

Custodians:
Navy - SH
Air Force - 99

Review Activities:
Air Force - 84
DLA - GS

User Activity:
Navy - MC

Preparing Activity:
DLA-IP

(Project 3436-0033)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-80085B	Positioners, Welding, Headstock and Tailstock Type

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
Defense Industrial Plant Equipment Center DIPEC-SSG	☐ VENDOR ☐ USER ☐ MANUFACTURER ☒ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)* Airways Blvd. Memphis, TN. 38114-5297	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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